

One Hit Wonders: recovering the longest period TESS planets

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ABSTRACT

TESS is searching for transiting planets around bright, nearby stars across nearly the entire sky. With a 27-day observing cadence, TESS will only be sensitive to close-in planets with periods of less than 14 days over the majority of the sky. We estimate TESS will observe a single-transit for approximately a thousand planets during its initial two-year mission. With unconstrained periods, these planets will be difficult to follow-up. The One Hit Wonders surveys will use an autonomous half-meter telescope to observe subsequent transits of the TESS single-transit planets in the North. Each night, up to a dozen targets with expected transits will be monitored at long-cadence; real-time data reduction will detect potential transits and trigger short-cadence, continuous observations. Simulations suggest approximately twenty planets may be recovered per observing year. Once confirmed, these planets will populate a previously sparse parameter space: long-period planets transiting bright host stars. As a bridge between hot Jupiters and the solar-system gas giants, studying the atmospheres of these planets may provide insight into the processes that sculpt planetary systems.

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1. INTRODUCTION

The *Kepler* mission showed that planets are ubiquitous in the galaxy (Howard et al. 2012; Dressing & Charbonneau 2013). In general, the planets that *Kepler* discovered were distant and faint, making it impossible to confirm via radial velocity measurements or perform follow-up observations to characterize the planets in detail. In particular, to feasibly reach the sufficient signal-to-noise required to definitively detect atmospheric biosignatures necessitates relatively bright stars, typically with $V < 10$ (Seager & Deming 2010). Lastly, while *Kepler*, over its prime mission and its extended K2 mission, provided exquisite photometry over long baselines, its sky coverage was limited, with only 1/400th of the sky monitored at a time.

The *Transiting Exoplanet Survey Satellite* (Ricker et al. 2014, TESS) was launched in Spring 2018 to search nearly the entire sky for transiting planets around nearby stars¹ To achieve such high sky coverage, TESS

tiles the sky into 26 sectors, dithering every 27.4 days to a new pointing. Thus over the majority of the sky (63% of the entire sky and 74% of the primary mission sky coverage) only planets with periods of less than 14 days will have multiple transits observed. Planets with longer periods may transit only a single time during the TESS observations, but, without the subsequent transits needed to constrain their ephemerides, following-up these planets is will be resource intensive.

With millions of stars being observed over a relatively short-baseline, more than a thousand single-transit planets are expected to be detected by TESS (Villanueva et al. 2018), over two orders-of-magnitude more than *Kepler* detected (Foreman-Mackey et al. 2016; Herman et al. 2019). Recovery of these planets is resource intensive, however there is significant scientific value in these systems. Almost all of the habitable-zone planets TESS will detect around M4 and earlier stars will have only a single observed transit (Kopparapu et al. 2013). As a bridge between the majority of TESS planets and solar system planets, the single-transit systems will also provide insight into open planetary formation questions, such as the migration mechanisms for hot Jupiters and

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¹ TESS has detected 471 planet candidates as of August 6, 2026.

planet-planet interactions (Cloutier & Lin 2013; Ziegler et al. 2018a).

The most important potential contribution of the recovered TESS single-transit planets is that, due to their long-periods and bright host stars, they will be excellent targets for atmospheric characterization. High-resolution transmission spectroscopy depends on bright stars to deliver sufficient signal from the molecular features in the exoplanet atmosphere (Birkby 2018). Only seven known systems host planets with orbital periods greater than 20 days transit stars brighter than 10th magnitude, all detected from space-based missions (five by *Kepler* and two by TESS). Planets with periods greater than a few days are difficult to detect from the ground (Pepper et al. 2003). A comprehensive model of exoplanet atmospheres requires probing a range of equilibrium temperatures; however, most atmospheric studies have been performed on hot Jupiters, and little is known about the atmospheres of cool exoplanets that are similar to the solar-system gas giants. This is in part due to a lack of suitable targets: scheduling ground-based observations of a full transit for an individual long-period planet can be difficult. With a large number of potential targets the probability that over a given time period a transit will be observable will significantly increase.

In this paper, we describe *One Hit Wonders*, a survey to recover the single-transit TESS planets. We begin in Section 2 by describing the simulated single-transit TESS yield and the feasibility of ground-based follow-up observations from a single site. In Section 3 we describe the survey hardware and strategy, and provide simulated survey yields. We conclude in Section 6.

2. SINGLE-SITE RECOVERY OF TESS SINGLE-TRANSIT PLANETS

In this section, we attempt to quantify the efficacy of performing ground-based follow-ups of the TESS single-transit planets from a single site. We begin by simulating the population of single-transit planets TESS will observe, and next estimate the number of these planets that will be observable from the ground.

2.1. Estimate of TESS single-transit yield

We use the TESS Input Catalog v7 (Stassun et al. 2018, TIC) for the population of planet host stars. The TIC provides stellar parameters, including TESS magnitude, stellar mass, radius, effective temperature, and coordinates, for each of 3.8 million stars. An estimate of the spectral type for each star was made using the temperature thresholds of (Pecaut & Mamajek 2013). The TESS observing baseline for each stars is equal to 27.4

days multiplied by the number of observed sectors for each star, which was determined using the known TESS pointings.

The host stars were populated with planets using the occurrence rates of Fressin et al. (2013) for AFGK-type stars and Dressing & Charbonneau (2013) for M-type stars. Each reference provides the number of planets around each star as a function of orbital period and planetary radius. Each star had planets populated at random using these occurrence rates.

The orbital parameters of each planet was estimated using the methods of Cooke et al. (2018). In brief, the orbital separation was calculated using Kepler’s third law, periastron angle as drawn from a uniform distribution between $-\pi$ and π , inclination stored as $\cos i$ and was drawn from a uniform distribution between 0 and 1, and eccentricity was drawn from a beta distribution with $\alpha=1.03$ and $\beta=13.6$ found by Van Eylen & Albrecht (2015). Transit durations (T_{dur}) were calculated using the relation in (Barclay et al. 2018),

$$T_{dur} = \frac{P}{\pi} \times \arcsin \left(\frac{R_{\star}}{a} \times \frac{\sqrt{1 + \frac{r_p}{R_{\star}} - b^2}}{\sqrt{1 - \cos^2 i}} \right). \quad (1)$$

where P is the orbital period, $R_{\star}$ is the planetary radius, a is the semimajor axis, and r_p is the planetary radius. The probability of transit was determined from the impact parameter using the relation from Winn (2010),

$$b = \frac{a \cos i}{R_{\star}} \times \frac{1 - e^2}{1 + e \sin \omega}. \quad (2)$$

where a transit is defined as occurring for $|b| < 1$.

For transiting planets, an initial transit time T_0 was drawn randomly from a uniform distribution between 0 and P . The planet transits during the TESS observation if $T_0 + n \times P$, where n is an integer, occurs within the TESS observation baseline of the host star. The number of subsequent transit observations by TESS can also be determined using P . To determine whether TESS can detect the transit, we estimate the S/N using the equation of Barclay et al. (2018),

$$S/N = \frac{\delta_{eff}}{\sigma_{1hr}} \sqrt{\frac{T_{dur}}{\Delta T}} \sqrt{n} \quad (3)$$

where δ_{eff} is effective transit depth, σ_{1hr} is the photometric precision in 1 hour of TESS data, n is number of observed transits, T_{dur} is transit duration in hours and ΔT is observing cadence. We estimate the effective transit depth using the equation,

$$\delta_{eff} = \left(\frac{R_p}{R_{\star}} \right)^2 \times \frac{1}{1 + C}, \quad (4)$$

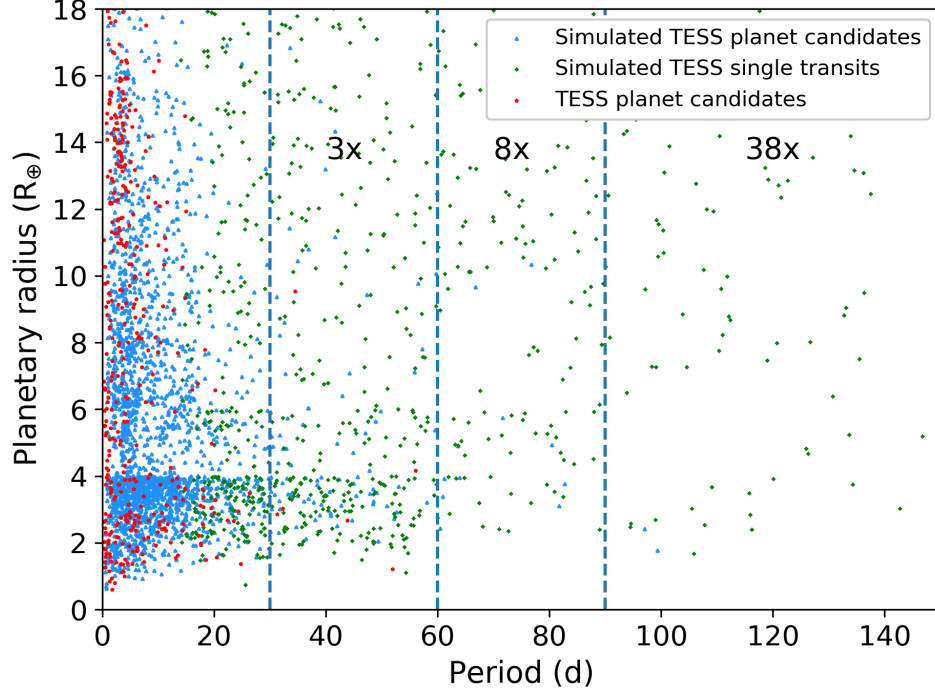


Figure 1. The orbital period-planet radius parameter space for the simulated yield of single-transit TESS planets compared to the multi-transit TESS planet simulation of Barclay et al. (2018). Also plotted are the released TOIs as of August 6, 2026. The large number of planets detected with low-radii is due to the selection criteria for the high-priority, postage stamp targets, in particular prioritizing M-dwarfs with detectable rocky planets. The potential enrichment in the yield of TESS planets for three period bins that would be achieved with full recovery of the observed single transit planets is also displayed.

where C is the contamination ratio from background stars and provided in the TIC. The photometric precision for TESS was estimated using the relations provided in Stassun et al. (2018). A transit was determined to be detected if the $S/N > 7.3$, a figure consistent with the literature (Sullivan et al. 2015; Cooke et al. 2018; Barclay et al. 2018).

We find that TESS will detect 904 ± 65 single-transit planet, determined from the the mean and standard deviation of multiple simulation runs. Approximately a quarter of these will come from the two-minute postage stamps (reserved for the 200,000 high priority targets), with the remainder from the thirty-minute full frame images. The total single-transit yield is comparable to that of two similar studies (Villanueva et al. 2018; Cooke et al. 2018).

The properties of single-transit planets is shown in comparison to the multi-transit TESS simulated yield in Figure 1. The single transit planets include the vast majority of TESS planets with periods greater than 30 days.

2.2. Single-site Observations

With a population of single-transit planets, we can determine the number that would be detectable from

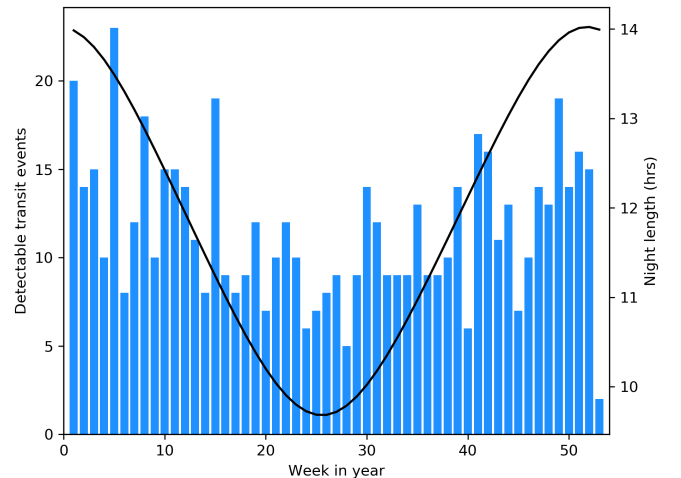


Figure 2. The number of detectable transit events from the TESS single-transit planets at a single, mid-latitude site, binned by week in the year. The length of the night is also displayed. On almost every day in the year, a transit of a TESS single-transit planet is detectable.

a single-site. We use the initial transit times, T_0 , and periods of the single-transit planets to determine future transits over a single year of observing. We consider a

transit potentially detectable if either the ingress, egress, or full transit occurs during the night.

We find that at a mid-latitude site (35° N), 615 ± 35 subsequent transits from the single-transit TESS planets are detectable. The detectable transits are well spread throughout the year regardless of night length, and on almost every night, at least one transit is detectable (see Figure 2). Without any foreknowledge on when these transits will occur, due to lack of ephemerides typically derived from multiple transits, blindly catching more than a few of these transits is infeasible.

3. THE ONE HIT WONDERS SURVEY

In this section, we describe the One Hit Wonders survey to recover the single-transit TESS planets. We begin by describing the hardware of the survey. We then discuss our survey strategy, including a method to estimate the period of the single-transit planets and vetting the targets for potential false positives. Finally, we present a simulation of our survey yield and discuss follow-up opportunities for these planets.

3.1. Survey Hardware

One Hit Wonders survey is located at Deep Sky West observatory near Rowe, NM, USA, and consists of a robotically operated 0.5-m telescope. The site is located at 2200m elevation, typically has 250 photometric nights a year, and an average sky darkness ≥ 21.7 mag arcsec $^{-2}$ in V-band. The optical tube assembly of the telescope is a PlaneWave Instruments CDK-20 with a corrected-Dall-Kirkham design. The f/6.8 output beam is reduced by further optics to f/4.5, resulting in a ~ 0.8 -square-degree field-of-view. The telescope is mounted on an Astro-Physics 3600GTO German Equatorial Mount. A 16Mpix Apogee U16M camera images the field at a plate scale of $0.82''\text{pix}^{-1}$. A filter wheel will allow observations through any of the Sloan filters; the choice of filter for each spectral type is chosen to maximize throughput based on the bandpass and the quantum efficiency of the camera and optics. The majority of the hardware used for One Hit Wonders was formerly used as the Dunlap Institute Arctic Telescope (Law et al. 2012).

A new addition is the use of an engineered diffuser in the filter wheel to enable precision photometry (Stefansson et al. 2017). Typically photometry on bright stars will be done by defocusing the telescope, spreading the PSF over a large area on the CCD. This allows longer exposures before saturation, increasing the overall duty cycle, and smooths out residual inter-pixel variations in pixel response after flat-fielding. Bright speckles remain, however, in the defocused images due to atmospheric seeing and telescope aberrations. A diffuser delivers an

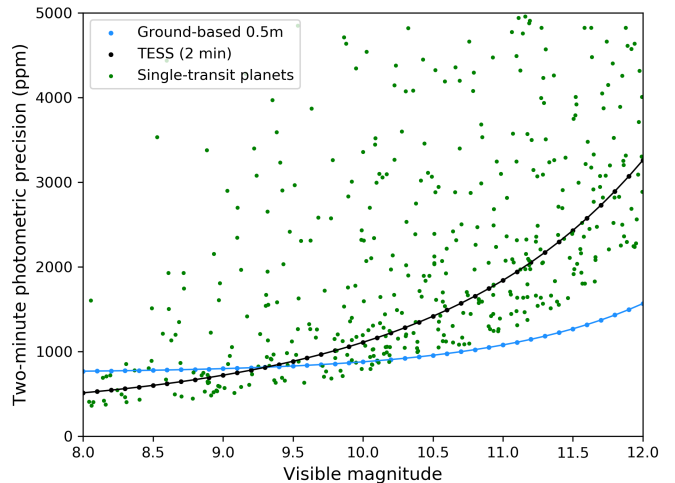


Figure 3. The photometric precision as a function of target magnitude for 2-minute exposures of TESS, from the predictions of Sullivan et al. (2015), and a ground-based diffuser-equipped 0.5-m telescope, estimated using *idiffuse* (Stefansson et al. 2018). The precision of the 0.5-m telescope is largely scintillation limited in the brightness regime shown, while TESS is primarily photon-limited. The transit depths for the simulated TESS single-transit planets from Section 2 are included for reference.

in-focus top-hat image of the target star, resulting in a photometric precisions on sub-meter telescopes superior to that of TESS (see Figure 3). We estimate the performance of a diffuser on a 0.5-m telescope using *idiffuse* (Stefansson et al. 2018)².

The telescope is controlled using a custom Python control software to perform scripted observations. The Python script interfaces with the hardware and proprietary software using ASCOM drivers and manufacturer’s drivers. Diffraction Limited’s Maxim-DL is used for CCD-camera control, image calibrations, and fine guiding, Software Bisque’s TheSkyX is used for pointing, and PinPoint Astrometric Engine 6.1, which interfaces with the astrometry.net cloud solving service, is used for plate solving.

3.2. Survey Strategy

The One Hit Wonders survey strategy uses the automated “real-time trigger” mode of operation proven by the MEarth survey (Berta et al. 2012) to monitor multiple targets and secure any transit detected. Each single-transit planet detected by TESS will have its orbital period and other planetary properties estimated from the TESS light curve and stellar parameters (as described in the subsequent section). These period estimates will

² Publicly available at <https://github.com/gummiks/idiffuse>

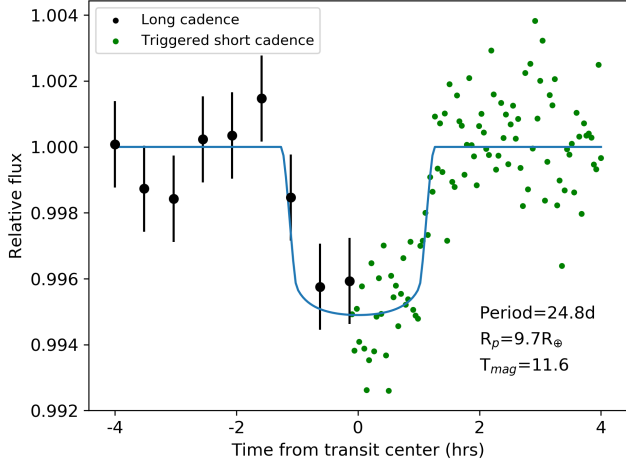


Figure 4. Simulation of a triggered light curve detected in the One Hit Wonders survey. The relative, normalized flux values from long cadence observations are shown with the estimated photometric precision. When consecutive outliers are detected, signaling a potential transit, continuous, short-cadence observations are triggered to secure the transit. A transit model, whose properties are displayed, can then be fit to the data, and follow-up observations scheduled to confirm and further characterize the planet.

provide some predictive advantage for scheduling the optimal observing times to increase the probability of observing a second transit. Each night, a target list will be assembled from the set of single-transit planets that are both observable and have potential transits based on the period estimates. The maximum number of targets observed in a given night is determined dynamically to optimize the detection efficiency, based on the duration and depth of the single transits in the TESS light curves. Typically, the number of targets observed a night is 6-8, and 12 targets is the hard limit.

The targets are observed at long-cadence, cycling between targets in an order that is optimized to reduce slew time. The exposure time of each target is determined to meet a precision threshold estimated based on the magnitude of the target star. On-the-fly data reduction and calibration is performed on the images. Aperture photometry is then performed using the Astropy affiliated package, Photutils, with bright (but not saturated) and nearby reference stars drawn from the TIC. Variable sources flagged by KELT (Oelkers et al. 2018), ASAS-SN (Jayasinghe et al. 2018), or Gaia DR2 (Holl et al. 2018) are excluded as reference stars. If the relative flux for a target from multiple in-progress observations all cross a significance threshold, subsequent high-cadence observations are triggered to confirm the transit event (see Figure 4). With the transit secured, follow-up observations, now easily scheduled with a constrained

period from the second transit observation, used to confirm and characterize the detected planet is then outsourced to other telescopes (e.g., by other members of the TESS seeing-limited photometry working group).

3.2.1. Orbital Period Estimation

To estimate the orbital periods of the TESS single transit planets we use the code described in Osborn et al. (2016), “*Namaste*: An Mcmc Analysis of Single Transiting Exoplanets” (hereafter, *Namaste*³). Originally developed for single transit planets in K2 light curves, *Namaste* was slightly modified to perform a Bayesian transit fit for the single transit in the TESS light curve to determine the planetary orbital parameters. We briefly outline the procedure described in detail in Osborn et al. (2016) below.

If we set eccentricity to be zero (a reasonable assumption as close-in planets are typically on nearly circular orbits, Kipping (2013)), the planetary period (P_{circ}) can be estimated from the scaled transit velocity (v'), that is the velocity with respect to the stellar radius, and stellar density (ρ_*) using *Kepler’s* laws,

$$P_{\text{circ}} = \frac{8\pi^2 G}{3} \frac{\rho_*}{v'^3} = 2\pi \frac{g}{R_* v'^3} \quad (5)$$

The scaled transit velocity is given by

$$v' \equiv \frac{v_{pl}}{R_*} = \frac{2\sqrt{(1 + R_p/R_*)^2 - b^2}}{T_D} \quad (6)$$

where impact parameter (b), planet-to-star ratio (R_p/R_*) and transit duration (T_D) are all provided by careful transit modeling by *Namaste*.

The TESS light curves are retrieved from the Mikulski Archive for Space Telescopes using *lightcurve* (Bartensen et al. 2019) for targets with 2-minute cadence observations, and with *eleanor* (Feinstein et al. 2019) for targets detected in the TESS full frame images. The TESS light curves are detrended and flattened with a Gaussian Process regression using *celerite* (Foreman-Mackey et al. 2017). The transit modeling is performed using the transit routines of Ian Crossfield⁴ and *emcee*, a Python implementation of Markov Chain Monte Carlo sampling (Foreman-Mackey et al. 2013). As longer periods orbits are less likely due to decreasing transit probability, a linear prior on transit velocity was set to discourage long periods. The resulting posteriors and light curves of a transit fit to a TESS single transit are shown in Figure 5.

³ Publicly available at <https://github.com/hposborn/namaste>

⁴ <http://www.mit.edu/~iancross/python/transit.html>

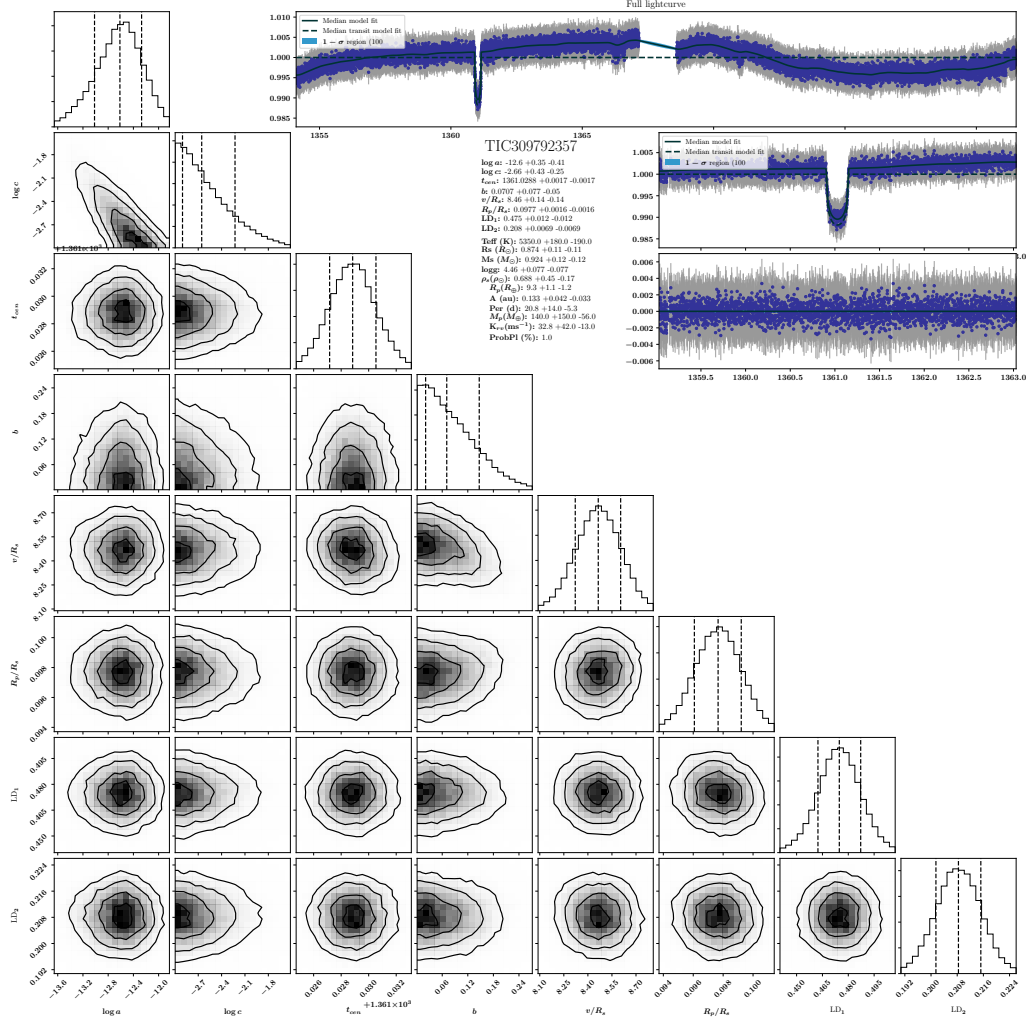


Figure 5. The posterior distribution of a transit fit by **Namaste** to a single transit detected by TESS around TIC309792357. The full raw TESS light curve is shown in the upper right with the Gaussian Process fit for detrending. Below that is the derived transit and planetary parameters, a zoom in centered on the single transit with the best-fit transit model, and residuals to the model.

For well constrained transit models, the largest source of uncertainty is the stellar parameters. For the stellar radius and effective temperature, we used the estimates and uncertainties reported in Gaia DR2 (Gaia Collaboration et al. 2018). For the estimated, stellar mass we used the values reported in the TIC (Stassun et al. 2018).

Need to test Namaste on multi-transit TESS planets to quantify period uncertainties...

3.2.2. Target Vetting

With a single-transit, many of the transit false positive tests typically implemented in data pipelines are not possible. The primary source of false positives in transit searches are astrophysical, such as blended background eclipsing binaries, transiting brown dwarfs, or grazing binaries. The tests used by TESS were devel-

oped for the *Kepler* mission (Batalha et al. 2013) and their effectiveness with single transit detections is described below.

A disparity in the depth of even and odd transits may signal a eclipsing binary, as two self-luminous bodies with different luminosities alternate eclipsing each other. This test is obviously not possible with single transits. Significant secondary transits, when the self-luminous eclipsing body orbits behind the host star, are searched for at a half-phase offset from the primary transit. These are often very shallow in visible light and detection is often beyond the photometric precision possible with TESS. In addition, without a second transit to set the phase, the search for even low-significance detections of secondary transits is difficult.

A V-shaped transit is more likely to be caused by a stellar binary than a planet. For stellar binaries, the two bodies are of near equal size, and the eclipsing body may never fully be encompassed by the stellar disk. For a transiting planet to produce a V-shaped transit it must be graze the stellar disk, only possible within a very particular range of possible impact parameters. In addition, a V-shaped planetary transit is difficult to model as a degeneracy exists between the impact parameter and the planetary radius.

The ephemeris of the transit will also be searched for possible matches with known transients in the field. Contamination from the TESS optics (“ghosting”), diffraction spikes, stellar crowding, and electronic cross-talk may result in a transit-like signal on the purported host star. The single-transit events are by definition non-periodic, “one-off” events and thus will not conform to known previously known transients such as variable stars. Thus the TESS pixel files must be individual checked for potential contamination. For example, TOI-225 was flagged as a single-transit candidate; further scrutiny revealed that the transit-like event was caused by the nearby passage of a previously unknown asteroid.

Lastly, a significant centroid offset in the point-spread function of the star during the transit may reveal a source offset from the target, such as a nearby eclipsing binary. However, the detection of small offsets, less than $\sim 10''$ (approximately half of a TESS pixel), with only a single transit is difficult.

To search for potential astrophysical false positive sources and unaccounted for photometric contamination, Robo-AO will be used to image in high-angular resolution each single-transit planet candidate. Robo-AO is a fully autonomous LGS-AO system mounted on the UH 2.2-m on Maunakea (Jensen-Clem et al. 2018), and has proven effective at identifying nearby stars to planet candidates (e.g., Ziegler et al. 2018b; Crossfield et al. 2016; Vanderburg et al. 2016). Possible nearby eclipsing binaries will also be searched for in the One Hit Wonders survey images.

4. SIMULATIONS OF SURVEY YIELD

We simulate our survey findings using the TESS single transit population in Section 2 and a Monte Carlo simulation with several underlying assumptions. We begin by cycling through each night of a simulated year of observing in New Mexico. For each night, we determine whether precision photometry observations are possible at random, with a 65% probability that a night is suitable (reasonable based on historical records for the site). We generate period estimates for the simulated single-

transit planets, drawn from a normal distribution centered on the true orbital period and with standard deviations equal to 15% of the true period. Uncertainties on the period estimates are also assumed to be 15%. Using the period estimates, the time ranges of potential future transits are propagated forward (the ranges increase in size as the uncertainties accrue) from the initial TESS transit time, T_0 . We assume that the observations are occurring concurrently with TESS observations and that planet candidates in each sector are delivered one month after the TESS observations end for that sector.

Then for each night in which observations can occur, we choose up to a dozen targets with predicted transits. We prioritize targets that are more likely to be detected due to their transit depth and host star magnitude, and for targets that have less subsequent transits since the TESS observations, as our period estimate for these targets has more predictive value. If the transit for one of these targets occurs during the night of observations, we estimate the number of long-cadence observations that would occur both in and out of transit. With our photometric precision estimate, described in Section 3.1, we can determine the significance of the transit detection and determine whether short-cadence observations would be triggered. If so, the planet is deemed “recovered”, and the rest of the night is ended under the assumption that the single target will be followed until sunrise. We add in several additional conservative factors to account for weather, false triggers, non-triggers. We do not account, however, for potential recovery of transits in later data reduction and analysis.

The above simulation yields an average of 24 ± 4 single transits recovered in the first year, with an average planetary radius of $6 \pm 3 R_{\oplus}$. Typically 2-4 habitable-zone planets are recovered. The survey yield in subsequent years taper off as the period estimates became less useful due to the uncertainties propagating. This does not account for the TESS extended mission, which will both detect additional transits to the single-transit planets from the primary mission, as well as detect new single-transit planets. We show the properties of the survey yield in Figure 6.

5. POTENTIAL FOR ATMOSPHERIC CHARACTERIZATION

The planets recovered in the simulations, having long orbital periods and hosting bright stars, fill in a previously sparse parameter space for known planets (see Figure 7). These features make them excellent targets for atmospheric studies, such as transmission spectroscopy. These observations require a bright host star to get suf-

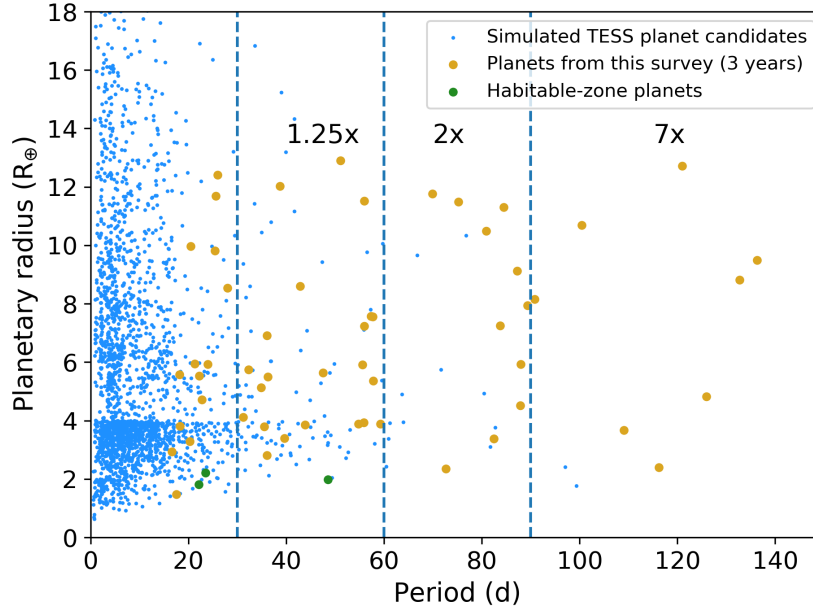


Figure 6. The properties of the simulated survey yield of One Hit Wonders over three years, including recovered habitable-zone planets. For comparison, the simulated TESS planet candidates of [Barclay et al. \(2018\)](#) are included. The enrichment in planet yield in three period bins to the TESS primary mission of One Hit Wonders is included for reference.

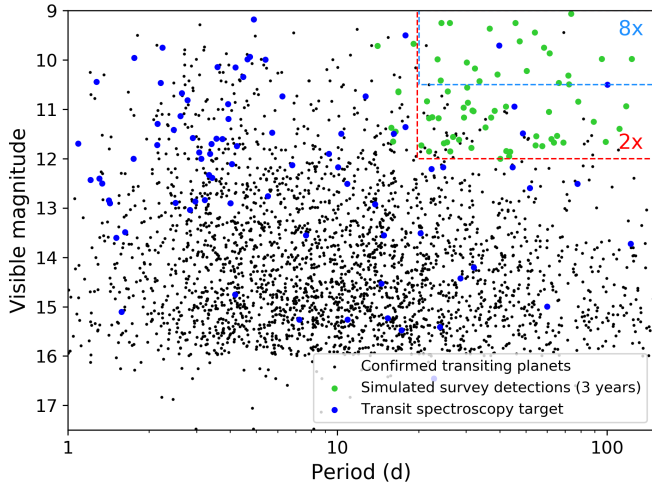


Figure 7. The period and visible magnitude of confirmed transiting exoplanets and the simulated survey detections from One Hit Wonders over three years. One Hit wonders can potentially vastly increase the number of long-period planets orbiting bright stars. The potential enrichment by the One Hit Wonders yield over currently known systems is shown in two bins. Planets which have had their atmospheres previously studied through transmission spectroscopy are also shown.

ficient signal-to-noise to detect molecular features in the exoplanet atmosphere.

Most atmospheric observations are on hot Jupiters ([Deming & Seager 2017](#)), orbiting at periods of a few days. These gaseous planets have equilibrium tempera-

tures in excess of 1000 K, with molecules such as TiO, VO and H₂O and refractory elements dominating the absorption spectrum ([Fortney et al. 2010](#)). At lower equilibrium temperatures, these molecules have condensed out of the upper atmosphere, allowing detection of molecules such as methane. These cool exoplanets will also act as a bridge between the hot Jupiters and the solar system gas giants, and may lead to a comprehensive model of planetary atmospheres at a range of equilibrium temperatures.

Hot Jupiters also often have condensate hazes ([Marley et al. 2013](#); [Kreidberg 2018](#)) which obscure all but the upper atmosphere. The source of these hazes is not understood, but there is evidence it may be scattering from small particles ([Dragomir et al. 2015](#)) or equilibrium condensates such as water, salt, sulfide or silicates ([Wakeford et al. 2017](#)). These particles will condense out and appear deeper in the atmosphere in planets with lower equilibrium temperatures. A lower, or non-existent, haze level will result in a larger portion of the atmosphere becoming translucent to star light during transit, raising the likelihood that increasingly complex molecules may be detected.

A radius gap was detected in the small planets detected by *Kepler* ([Fulton et al. 2017](#)). This radius “valley” can be interpreted as the transition region between rocky super-Earths and sub-Neptunes, with sufficient gravity to hold onto their primordial -thick gaseous envelope ([Owen & Wu 2017](#)). The H/He envelope of these

planets may evaporate during migration caused by perturbations from secondary stellar bodies (Ziegler et al. 2018a) or planet-planet scattering (Xie et al. 2016). Recently, helium in an eroding atmosphere of an exoplanet was detected (Spake et al. 2018). A few of the single-transit planets recovered by One Hit Wonders may orbit in a “sweet-spot”, with sufficient radiation pressure to evaporate their thick atmospheres but at timescales allowing detection past the protoplanetary stage. Studying the properties of these systems will be important tests for models of exoplanet formation and evolution.

6. CONCLUSION

TESS will find nearly a thousand single-transit planets during its prime mission due to its high-sky coverage and short observing baseline. These planets are scien-

tifically valuable, making excellent targets for studies of cool exoplanet atmospheres. However, without knowledge of the orbital ephemerides, scheduling follow-up observations is not possible. One Hit Wonders is using a half-meter robotic telescope, guided by estimates of the orbital periods derived from the stellar parameters and TESS light curves, to recover these planet by detecting a second transit. We simulate that 24 ± 4 of these single-transit planets may be detected by One Hit Wonders in the first year.

Facilities: TESS

Software: astropy, emcee, celerite, Namaste, idiffuse, eleanor, lightkurve

REFERENCES

- Barclay, T., Pepper, J., & Quintana, E. V. 2018, The Astrophysical Journal Supplement Series, 239, 2
- Barentsen, G., Hedges, C., Vincius, Z., et al. 2019, KeplerGO/lightkurve: Lightkurve v1.0b30, , , doi:10.5281/zenodo.2611871. <https://doi.org/10.5281/zenodo.2611871>
- Batalha, N. M., Rowe, J. F., Bryson, S. T., et al. 2013, ApJS, 204, 24
- Berta, Z. K., Irwin, J., Charbonneau, D., Burke, C. J., & Falco, E. E. 2012, AJ, 144, 145
- Birkby, J. L. 2018, arXiv e-prints, arXiv:1806.04617
- Cloutier, R., & Lin, M.-K. 2013, MNRAS, 434, 621
- Cooke, B. F., Pollacco, D., West, R., McCormac, J., & Wheatley, P. J. 2018, A&A, 619, A175
- Crossfield, I. J. M., Ciardi, D. R., Petigura, E. A., et al. 2016, The Astrophysical Journal Supplement Series, 226, 7
- Deming, D., & Seager, S. 2017, arXiv e-prints, arXiv:1701.00493
- Dragomir, D., Benneke, B., Pearson, K. A., et al. 2015, ApJ, 814, 102
- Dressing, C. D., & Charbonneau, D. 2013, ApJ, 767, 95
- Feinstein, A. D., Montet, B. T., Foreman-Mackey, D., et al. 2019, arXiv e-prints, arXiv:1903.09152
- Foreman-Mackey, D., Agol, E., Angus, R., & Ambikasaran, S. 2017, ArXiv. <https://arxiv.org/abs/1703.09710>
- Foreman-Mackey, D., Hogg, D. W., Lang, D., & Goodman, J. 2013, PASP, 125, 306
- Foreman-Mackey, D., Morton, T. D., Hogg, D. W., Agol, E., & Schölkopf, B. 2016, AJ, 152, 206
- Fortney, J. J., Shabram, M., Showman, A. P., et al. 2010, ApJ, 709, 1396
- Fressin, F., Torres, G., Charbonneau, D., et al. 2013, ApJ, 766, 81
- Fulton, B. J., Petigura, E. A., Howard, A. W., et al. 2017, AJ, 154, 109
- Gaia Collaboration, Brown, A. G. A., Vallenari, A., et al. 2018, A&A, 616, A1
- Herman, M. K., Zhu, W., & Wu, Y. 2019, arXiv e-prints, arXiv:1901.01974
- Holl, B., Audard, M., Nienartowicz, K., et al. 2018, A&A, 618, A30
- Howard, A. W., Marcy, G. W., Bryson, S. T., et al. 2012, ApJS, 201, 15
- Jayasinghe, T., Stanek, K. Z., Kochanek, C. S., et al. 2018, arXiv e-prints, arXiv:1809.07329
- Jensen-Clem, R., Duev, D. A., Riddle, R., et al. 2018, AJ, 155, 32
- Kipping, D. M. 2013, MNRAS, 434, L51
- Kopparapu, R. K., Ramirez, R., Kasting, J. F., et al. 2013, ApJ, 765, 131
- Kreidberg, L. 2018, Exoplanet Atmosphere Measurements from Transmission Spectroscopy and Other Planet Star Combined Light Observations, 100
- Law, N. M., Sivanandam, S., Murowinski, R., et al. 2012, in Society of Photo-Optical Instrumentation Engineers (SPIE) Conference Series, Vol. 8444, Ground-based and Airborne Telescopes IV, 84445C
- Marley, M. S., Ackerman, A. S., Cuzzi, J. N., & Kitzmann, D. 2013, Clouds and Hazes in Exoplanet Atmospheres, ed. S. J. Mackwell, A. A. Simon-Miller, J. W. Harder, & M. A. Bullock, 367

- Oelkers, R. J., Rodriguez, J. E., Stassun, K. G., et al. 2018, *AJ*, 155, 39
- Osborn, H. P., Armstrong, D. J., Brown, D. J. A., et al. 2016, *MNRAS*, 457, 2273
- Owen, J. E., & Wu, Y. 2017, *ApJ*, 847, 29
- Pecaut, M. J., & Mamajek, E. E. 2013, *The Astrophysical Journal Supplement Series*, 208, 9
- Pepper, J., Gould, A., & Depoy, D. L. 2003, *AcA*, 53, 213
- Ricker, G. R., Winn, J. N., Vanderspek, R., et al. 2014, in *Society of Photo-Optical Instrumentation Engineers (SPIE) Conference Series*, Vol. 9143, *Space Telescopes and Instrumentation 2014: Optical, Infrared, and Millimeter Wave*, 914320
- Seager, S., & Deming, D. 2010, *Annual Review of Astronomy and Astrophysics*, 48, 631
- Spake, J. J., Sing, D. K., Evans, T. M., et al. 2018, *Nature*, 557, 68
- Stassun, K. G., Oelkers, R. J., Pepper, J., et al. 2018, *AJ*, 156, 102
- Stefansson, G., Mahadevan, S., Hebb, L., et al. 2017, *ApJ*, 848, 9
- Stefansson, G., Mahadevan, S., Wisniewski, J., et al. 2018, in *Society of Photo-Optical Instrumentation Engineers (SPIE) Conference Series*, Vol. 10702, *Ground-based and Airborne Instrumentation for Astronomy VII*, 1070250
- Sullivan, P. W., Winn, J. N., Berta-Thompson, Z. K., et al. 2015, *ApJ*, 809, 77
- Van Eylen, V., & Albrecht, S. 2015, *ApJ*, 808, 126
- Vanderburg, A., Bieryla, A., Duev, D. A., et al. 2016, *ApJ*, 829, L9
- Villanueva, Steven, J., Dragomir, D., & Gaudi, B. S. 2018, *arXiv e-prints*, arXiv:1805.00956
- Wakeford, H. R., Visscher, C., Lewis, N. K., et al. 2017, *MNRAS*, 464, 4247
- Winn, J. N. 2010, *arXiv e-prints*, arXiv:1001.2010
- Xie, J.-W., Dong, S., Zhu, Z., et al. 2016, *Proceedings of the National Academy of Science*, 113, 11431
- Ziegler, C., Law, N. M., Baranec, C., et al. 2018a, *AJ*, 156, 83
- . 2018b, *AJ*, 155, 161

TIC ID	TOI	Actual Period (Days)	Estimated Period (Days)	Percentage Difference
388104525	112.01	2.49979	2.38	4.79
149603524	102.01	4.41195	4.51	2.22
144065872	105.01	2.18467	3.21	46.93
92352620	107.01	3.95005	2.79	29.37
230982885	195.01	2.07276	3.02	45.70
25375553	143.01	2.31089	2.30	0.47
267263253	135.01	4.12690	3.24	21.49
100100827	185.01	0.94145	1.92	103.94
183532609	191.01	8.15917	4.12	49.50
166836920	267.01	5.75244	4.62	19.69

Table 1. Namaste Multi-Transit Estimation